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What are Gamma Distributions?

A Physics Exposition

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PHYS 721

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Gamma Distribution

Let $X \sim \text{Gamma}(\alpha, \beta)$

Then,

$$P_X(x) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta} & 0 < x < \infty \\ 0 & \text{OW} \end{cases}$$

But,

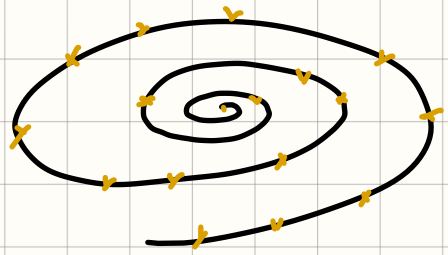
- Why does it have this form?
- What does it model?

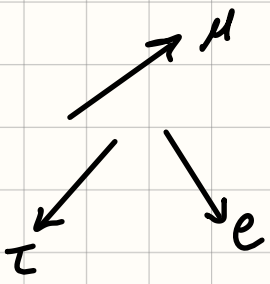
The Exposition

We are going to explore 3 interpretations.

1. Astro Particle Physics — "Particle Arrivals & Residuals"
2. Statstical Mechanics — "3D Ideal gas models"
3. Quantum Mechanics — "Hydrogen Atom"

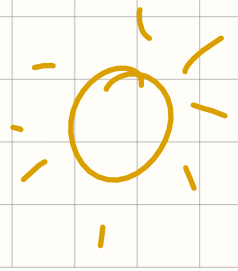






Astroparticle Physics





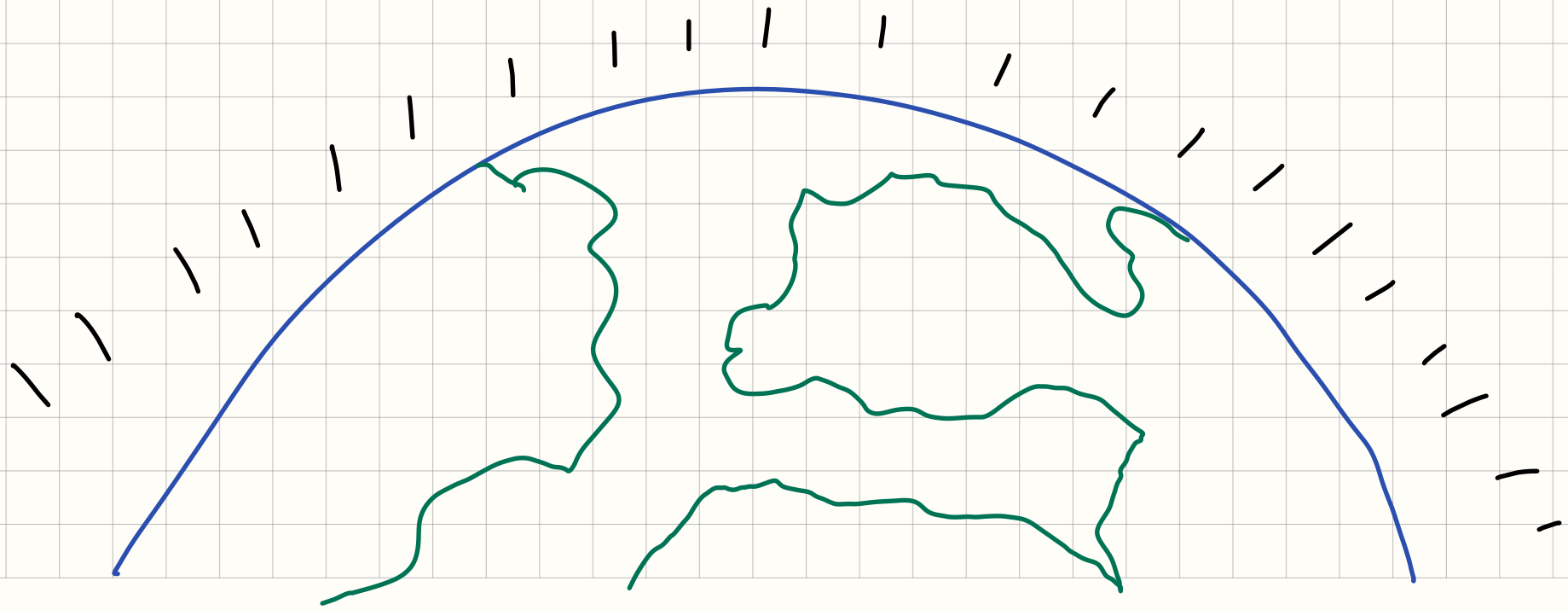




Arrival Times of Particles

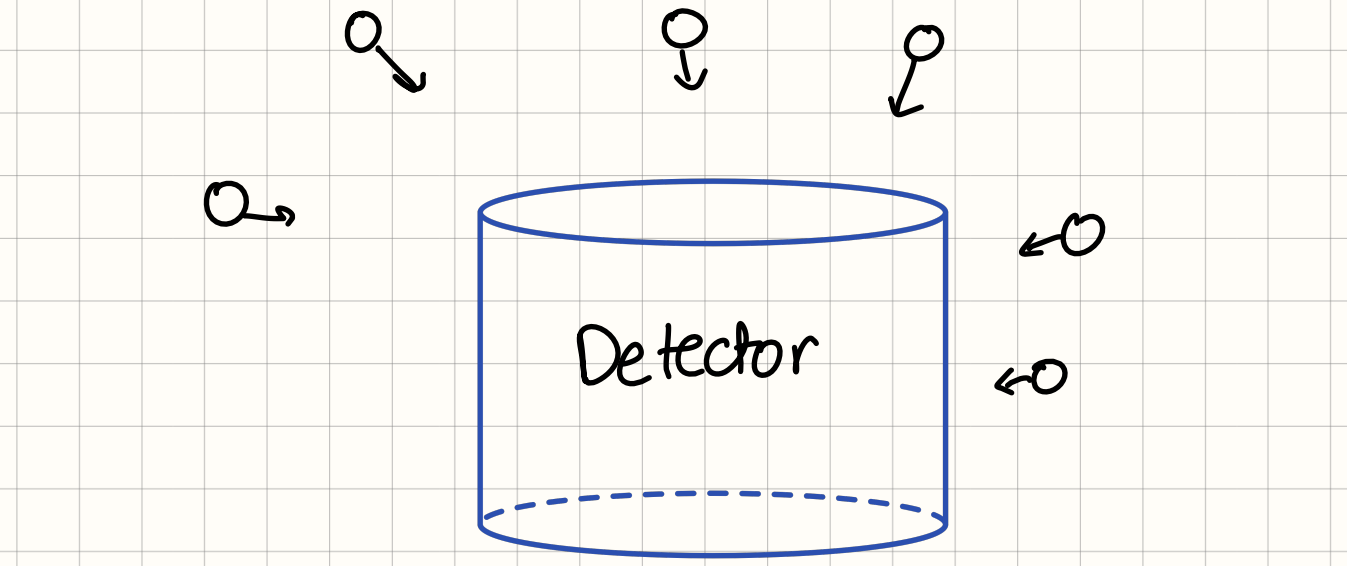
Neutrinos come from every where around us.

Learning about their rates and fluxe can give us an idea to what sources produced them & how



Poisson Process

What is the probability that the kth particle arrives?





Binomial Distribution

Flipping coins



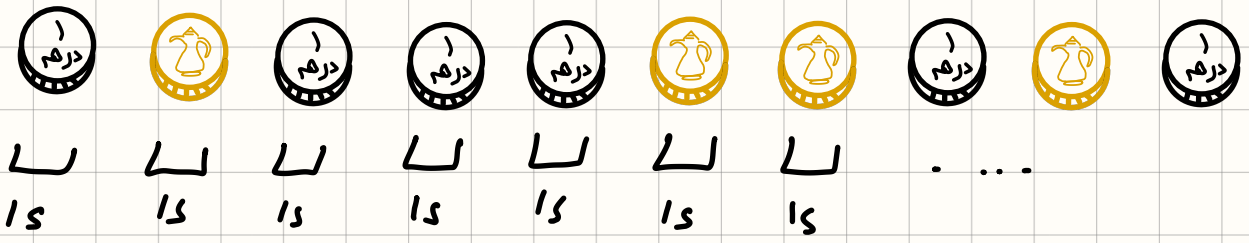
Whats the probability to get 4 dallas in 10 coins ?

$$P_X(k) = \begin{cases} \binom{n}{k} p^k (1-p)^{n-k} & 0 \le k \le n \\ 0 & \text{ow} \end{cases}$$

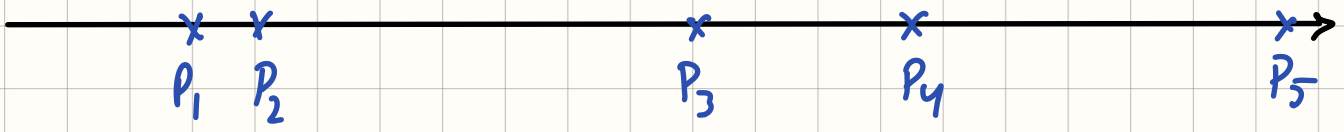
Binomial Distribution of $k=4$, $n=10$

Bernoulli Process

We can also think of our problem as a process.



discrete time



continuous time

What if we take our $n \to \infty$ but let our mean np stay constant?

The Math: Binomial to Poisson

Let $X \sim \text{Binomial}(n, p)$ s.t. $\mathbb{P}(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$

Impose $p = \lambda/n$ with $\lambda = np$ fixed. As $n \to \infty$,

$$\mathbb{P}(X=k) = \binom{n}{k}\left(\frac{\lambda}{n}\right)^k \left(1-\frac{\lambda}{n}\right)^{n-k}$$

$$= \frac{n(n-1)\dots(n-k+1)}{k!\ n^k} \lambda^k \left(1-\frac{\lambda}{n}\right)^n \left(1-\frac{\lambda}{n}\right)^{-k}$$

$$\to 1/k! \cdot \lambda^k \cdot e^{-\lambda} \cdot 1$$

$$\boxed{\mathbb{P}(X=k) \to \frac{\lambda^k}{k!} e^{-\lambda}}$$

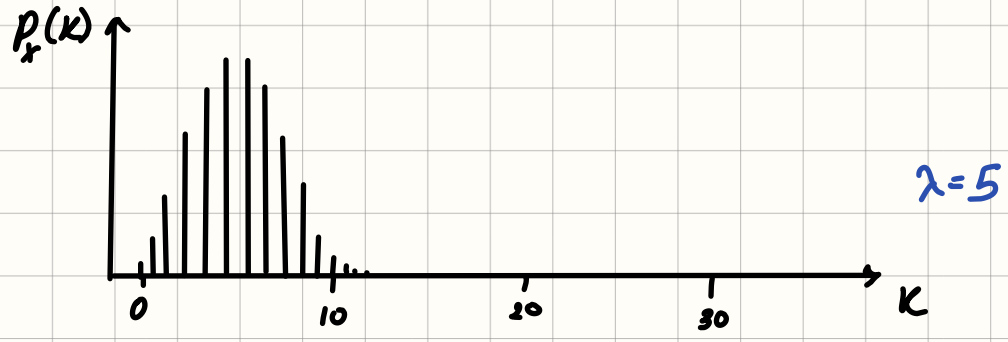
Poisson Distribution



Measures the probability that k particles with arrival rate λ and time J

$$P_X(k) = \begin{cases} \frac{(\lambda J)^k}{k!} e^{-\lambda J} & \text{for } k = 0, 1, 2 \dots \\ 0 & \text{OW} \end{cases}$$

Graph:



The Math: Poisson to Erlang



Let T_k be the time of the kth arrival

$$\{T_k \le t\} \leftrightarrow \{k \text{ arrivals or more by time } t\}$$

$$F_{T_k} = \mathbb{P}(T_k \le t) = \sum_{n=k}^{\infty} P(n,t) = 1 - \sum_{n=0}^{k-1} P(n,t) = 1 - \sum_{n=0}^{k-1} \frac{(\lambda t)^n e^{-\lambda t}}{n!}$$

Differentiating with respect to t:

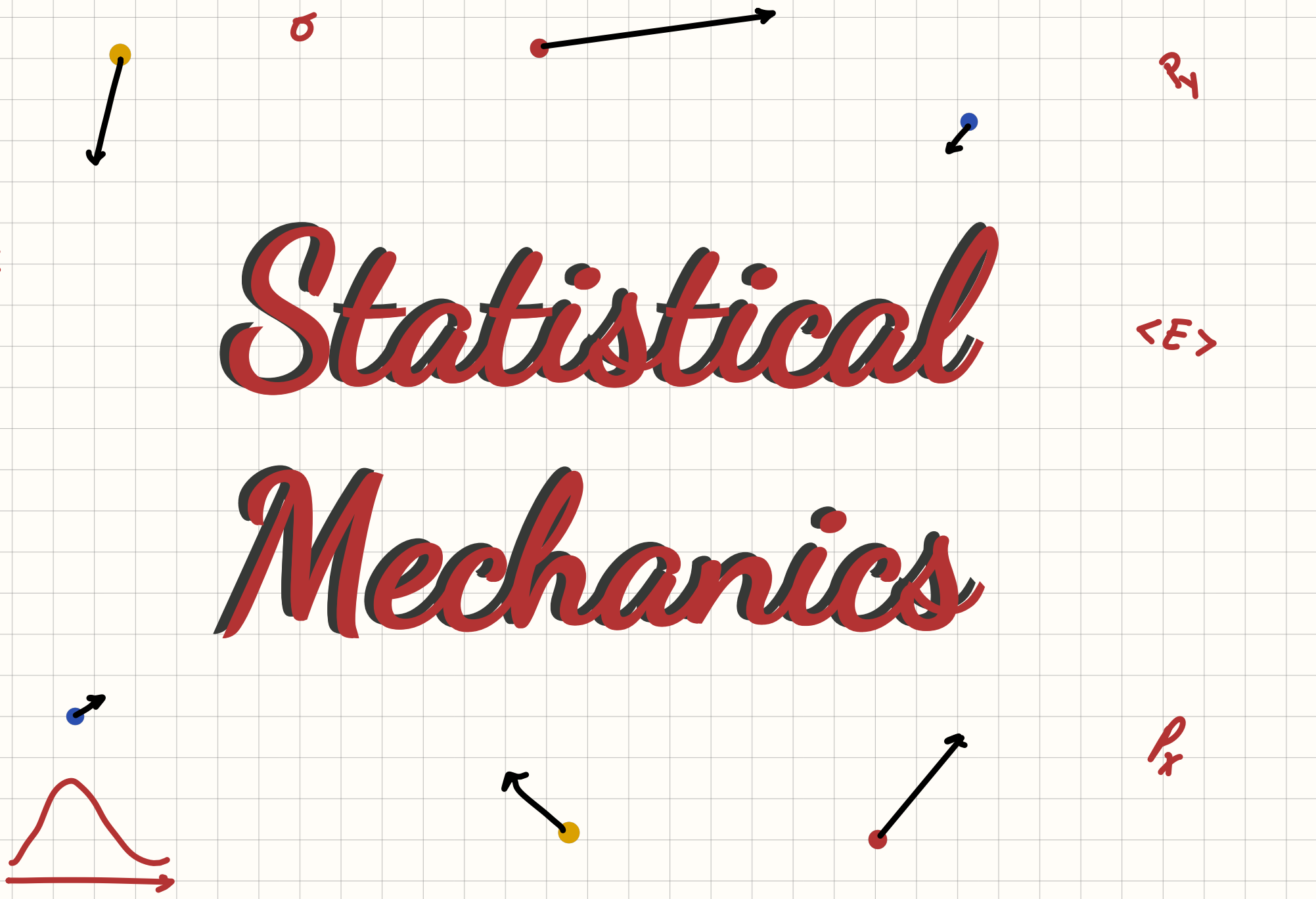
$$f_{T_k}(t) = \frac{d}{dt} F_{T_k}(t) = \frac{\lambda^k t^{k-1} e^{-\lambda t}}{(k-1)!} \qquad t \ge 0$$

This is an Erlang distribution. But, also Gamma $(k, 1/\lambda)$

k: integer

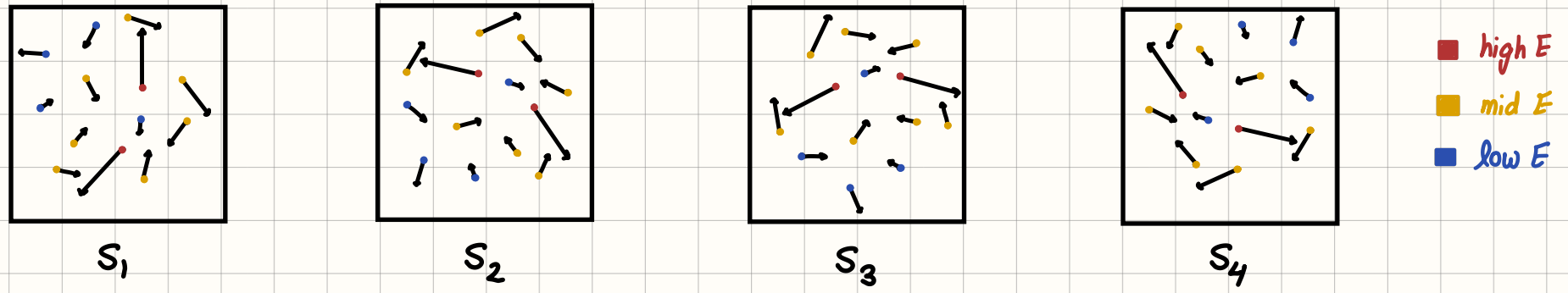
$1/\lambda$: reciprocal of arrival rate.

Statistical Mechanics



Energy Distribution of Ideal Gas

What is the energy distribution of a 3D ideal gas ?

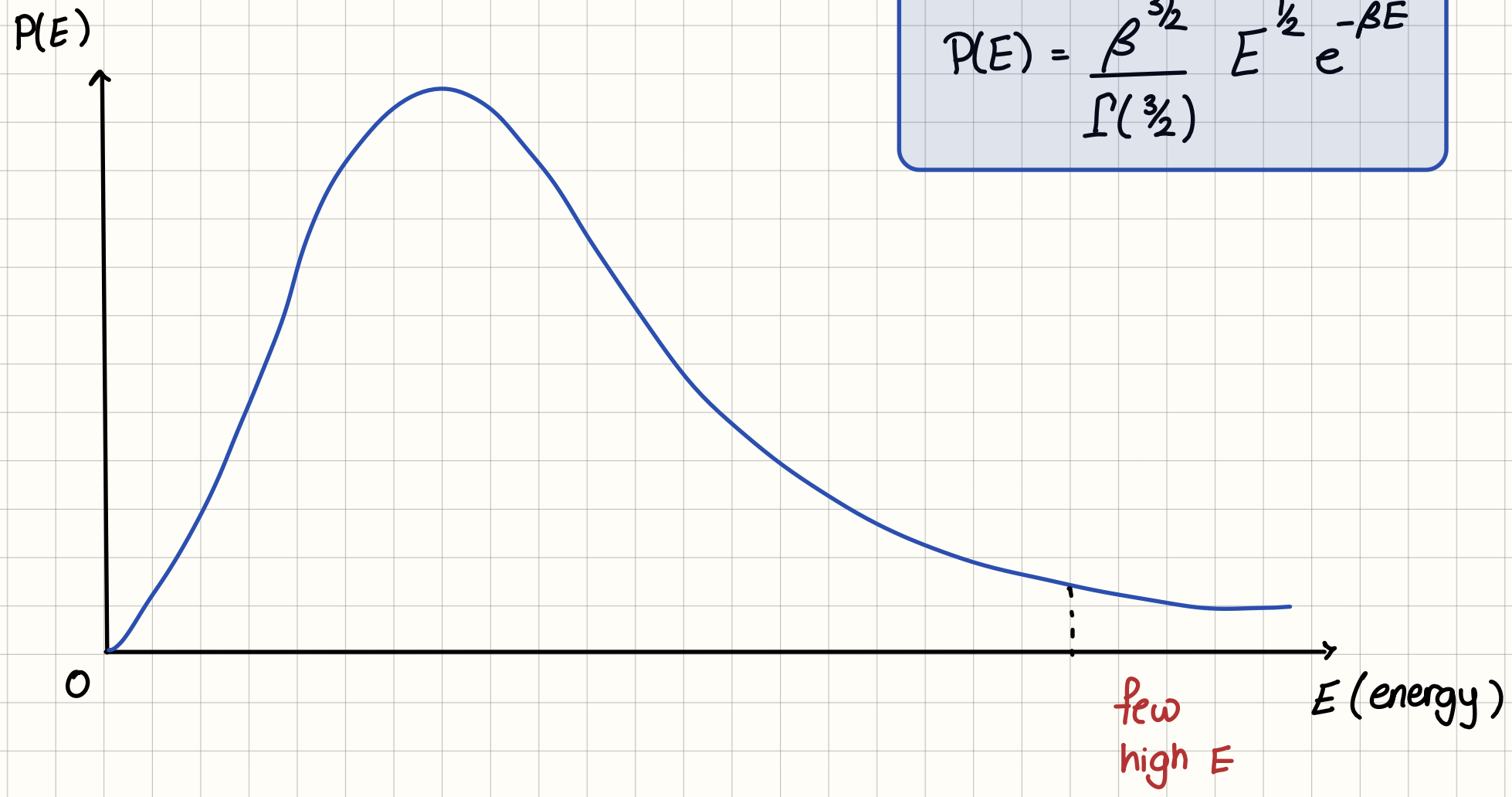


$$p(s) \propto \exp\{-\beta \mathcal{H}(s)\}$$

- $p(s)$ → (q_i, P_i)
  - q_i: positions, 3D
  - P_i: momenta, 3D
- $\exp\{-\beta \mathcal{H}(s)\}$ → Hamiltonian

Boltzmann Factor !

$$P(E) = \frac{\beta^{3/2}}{\Gamma(3/2)} E^{1/2} e^{-\beta E}$$



A Gamma again ?

The Math: Sum of Square of Gaussians

Suppose a particle is in thermal equilibrium,

$$p_x, p_y, p_z \sim N(0, mk_BT)$$

Normalize

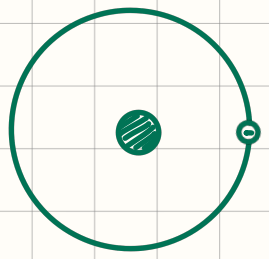
$$Z_i \sim \frac{p_i}{\sqrt{mk_BT}} \qquad i = 1, 2, 3$$

Now, $E = \frac{p_x^2 + p_y^2 + p_z^2}{2m} = \frac{k_BT}{2}\left(Z_1^2 + Z_2^2 + Z_3^2\right)$

from statistical theory, $Z_1^2 + Z_2^2 + Z_3^2 \sim \chi_3^2 \sim \text{Gamma}(3/2, 2)$

Scaling by $1/2\beta$,

$$E \sim \text{Gamma}(3/2, 1/\beta)$$



$e^{i\omega t}$?

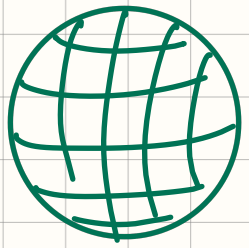
∇^2

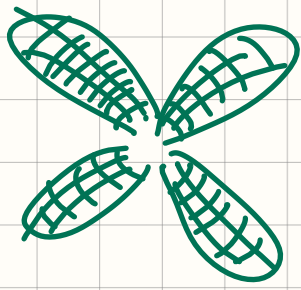
$\langle \varphi | \varphi \rangle$

$\uparrow\downarrow$

Quantum Mechanics

$a^\dagger a$





ψ



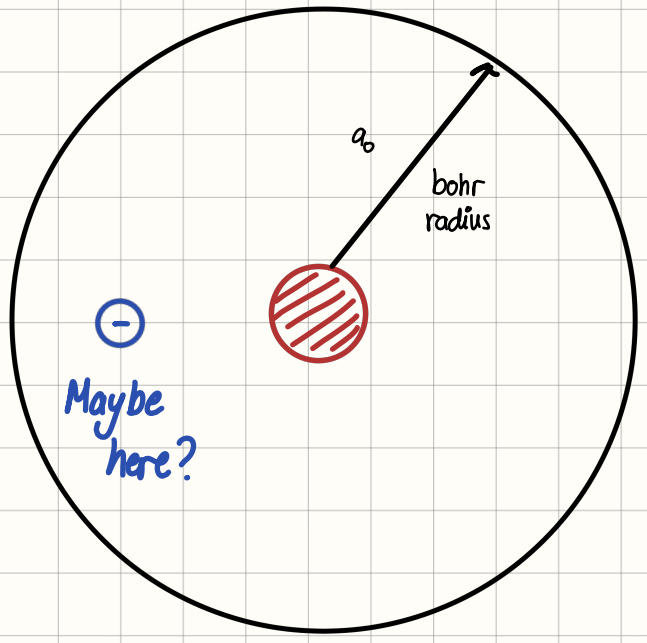
$E = \hbar\omega\left(a^\dagger a + \tfrac{1}{2}\right)$



$E = ?$

Electron Radial Distribution in Hydrogen Atom

What is the probability distribution of the radial distance of an electron in a hydrogen atom ?

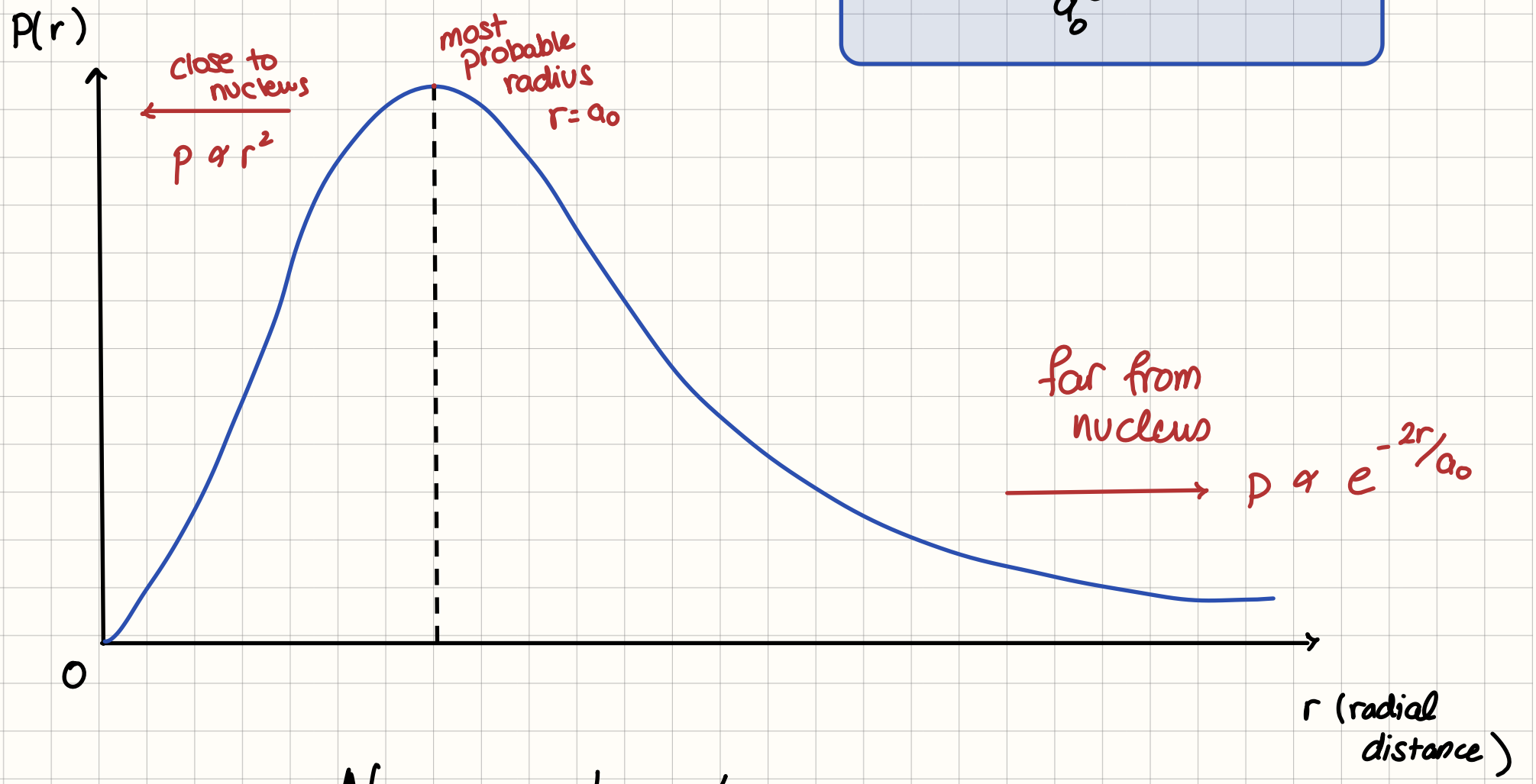


Is it here ?

hydrogen atom

Ground State

$$P(r) = \frac{4}{a_0^3} r^2 e^{-2r/a_0}$$



No way! I'm done...

The Math: Hydrogen Atom PDE

Schrödinger Equation for a hydrogen like atom

$$-\frac{\hbar^2}{2m}\nabla^2\psi - \frac{Ze^2}{4\pi\varepsilon_0 r}\psi = E\psi$$

Using seperation of variables and dimension less radial variable $\rho = qr$

$\chi(\rho) = \psi(r)$

$$\frac{d}{d\rho^2}(\rho\chi) + \left(\frac{\lambda}{\rho} - \frac{1}{4} - \frac{L(L+1)}{\rho^2}\right)\rho\chi = 0$$

↳ This turns out to be an **Associated Laguerre Equation**

The Math: Gamma & Associated Laguerre

The radial solution is the Laguerre Functions

$$\rho\chi(\rho) = \underbrace{e^{-\rho/2}\rho^{L+1}}_{\text{Gamma Kernel!}} \underbrace{L^{2L+1}_{\lambda-L-1}(\rho)}_{\text{Associated Laguerre Polynomials}}$$

Turns out that the Gamma Kernel is the weight function to make associated laguerre equations self adjoint!

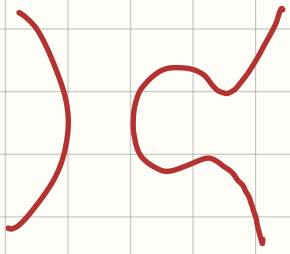
$$w_k(x) = \frac{1}{x}\exp\left[\int \frac{k+1-x}{x}\,dx\right] = x^k e^{-x}$$

Fermat's Last Theorem



$$x^n + y^n \overset{?}{=} z^n, \quad n \geq 3$$





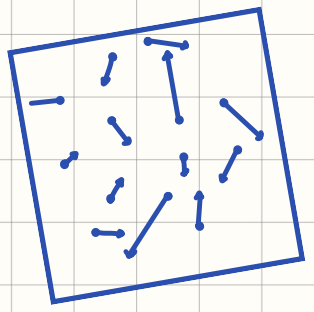
Elliptic Curves





Modular Forms









Thank You



